METRIC

DOD-B-24506/2C(SH)
20 March 1978
SUPERSEDING
DOD-B-24506/2B(SH)
1 June 1976

MILITARY SPECIFICATION SHEET

BATTERY, STORAGE, SILVER-ZINC ALKALINE TYPE

(FOR TRIESTE) AUXILIARY (5000 AMPERE-HOUR) (METRIC)

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the battery described herein shall consist of this document and the latest issue of DOD-B-24506.

REQUIREMENTS:
Electrical characteristics: See table I.

TABLE I - Electrical characteristics.

Discharge rate ampere (A)	Time hours (h)	Capacity ampere-hour (Ah)	Battery minimum voltage (V)	Minimum average[1/] cell voltage (V)	Minimum (cut-off) cell voltage (V)
2140	.2	4280	20	1.44	1.20
1170	4	4680	21	1.46	1.25
625	8	5000	21.5	1.48	1.30
325	16	5200	22	1.50	1.34
260	20	5200	23	1.52	1.36

1/ The "average voltage" is defined as the average of voltage recordings taken at equal intervals, starting with the first recording (taken between 1/2 minute and 1 minute after applying a constant rate) and including the last recording. The interval of readings shall be not greater than h/5, where h is the time in hours of the rated capacity of the test. The time to the last recording shall be not greater than the rated time.

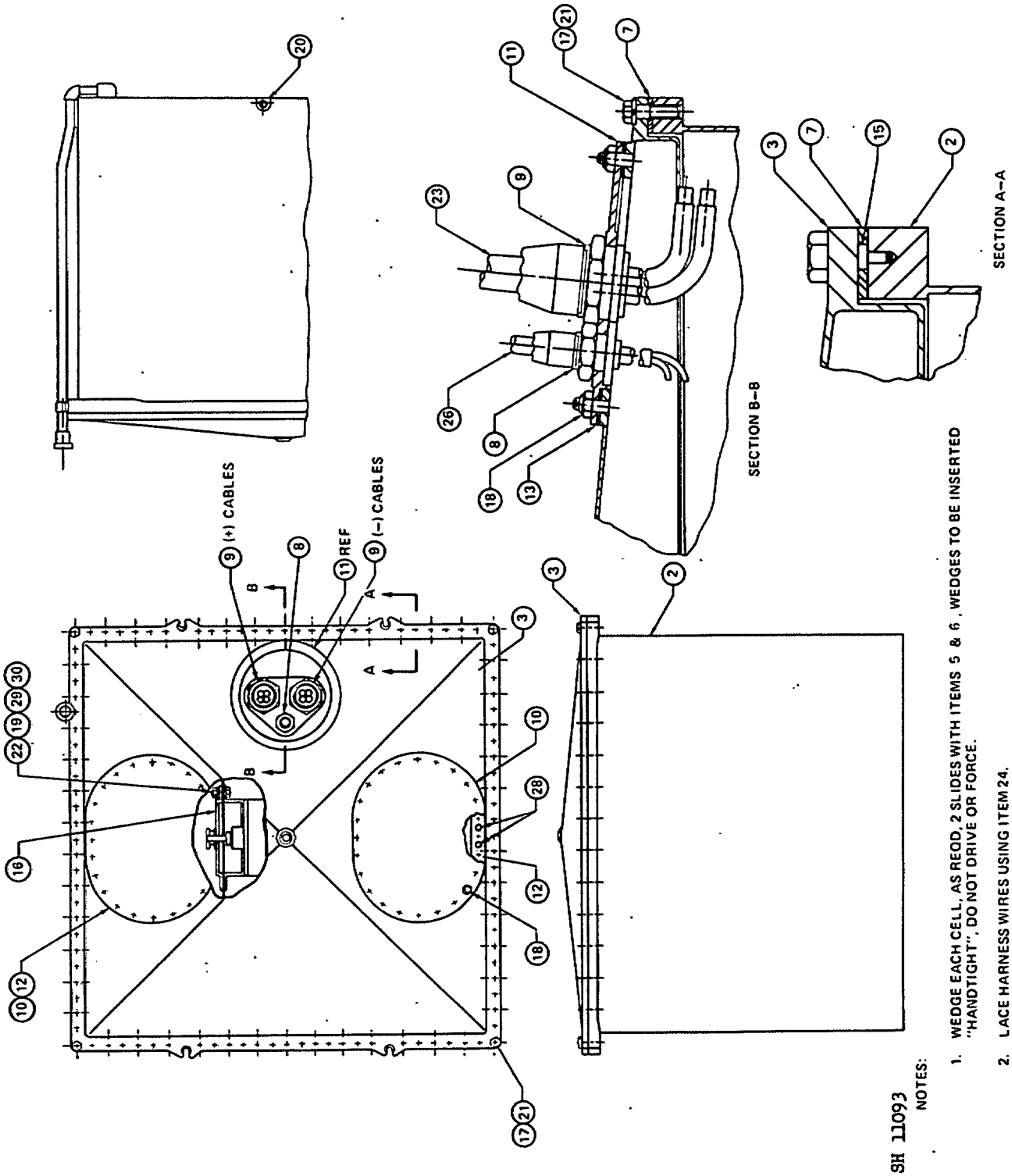
Life: 24 months or 400,000 Ah, whichever occurs first.
Cell dimensions - millimeters (mm):
Maximum length: 273 mm.
Maximum width: 273 mm.
Maximum height: (including electrolyte entrainment eliminator: 772 mm.
Cell mass: 102 kilograms (kg) (maximum); 100 kg (average).
Battery assembly: See figures 1, 2, 3, and 4.
© Total cells offered for delivery: 24.
(a) First article cells: 2.
(b) Battery cells: 16.
(c) Spare cells: 4.
(d) Sample cells for quality conformance: 2.
K factor for quality conformance (capacity): 5.3.
Hydrogen evolution: maximum 100 ml/hour.
© Charge efficiency: For determinations of charge efficiency curve.
(a) All discharges at the 8-hr rate
(b) Charges at the 50, 100, 150, 200, 300, and 400-ampere rates.

Preparing activity:
Navy - SH
(Project 6140-N474-2)

© denotes changes

FSC 6140

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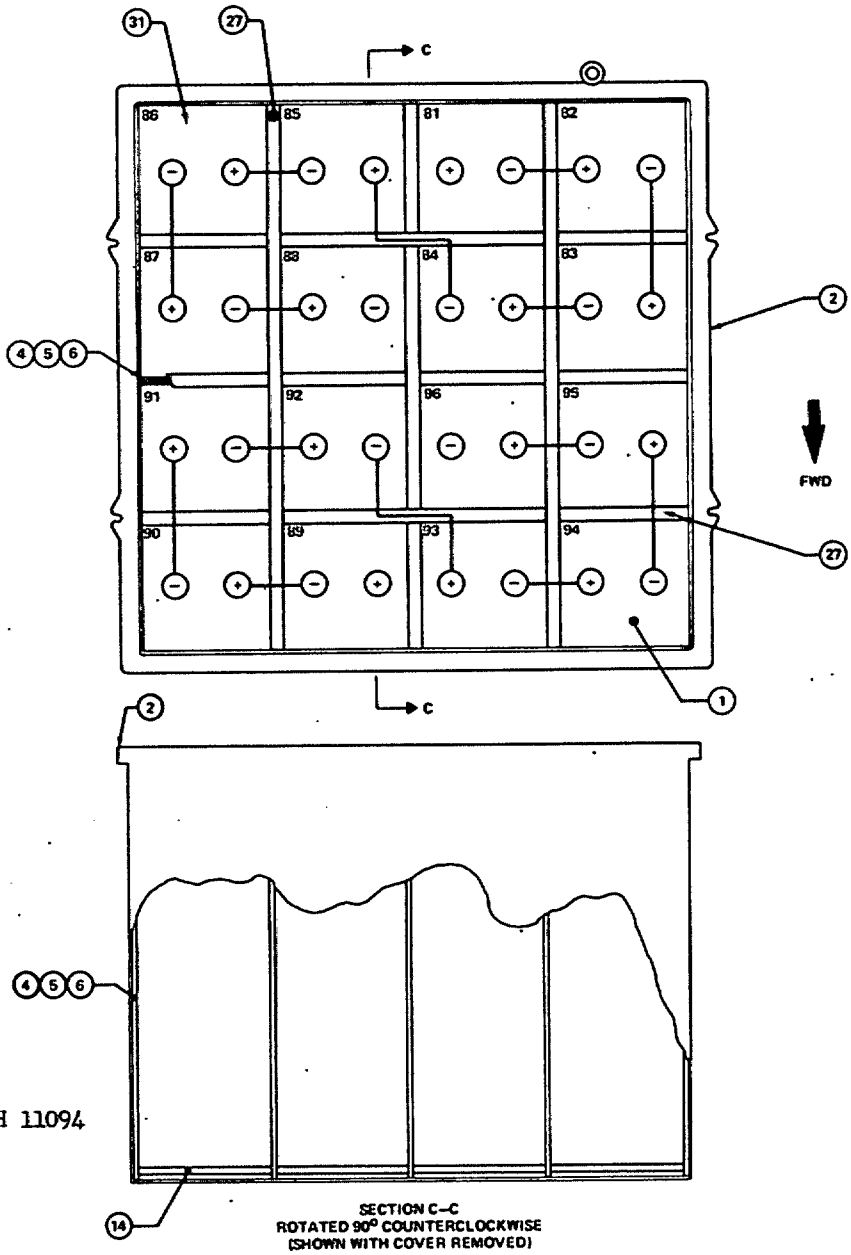


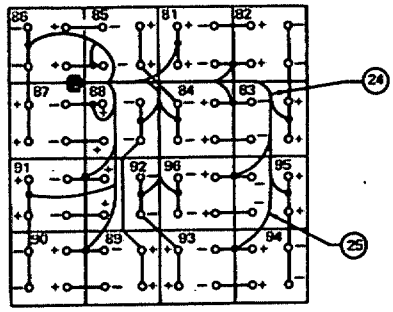
NOTES:

1. WEDGE EACH CELL, AS REQD, 2 SLIDES WITH ITEMS 5 & 6, WEDGES TO BE INSERTED "HANDTIGHT", DO NOT DRIVE OR FORCE.
2. LACE HARNESS WIRES USING ITEM 24.

FIGURE 1. Battery assembly (battery box and cover).

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MONITOR HARNESS
WIRING DIAGRAM

NOTE: ICV wiring and intercell connection layout is typical; changes may be made as necessary for optimum design.

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FIGURE 2. Battery assembly (typical layout).

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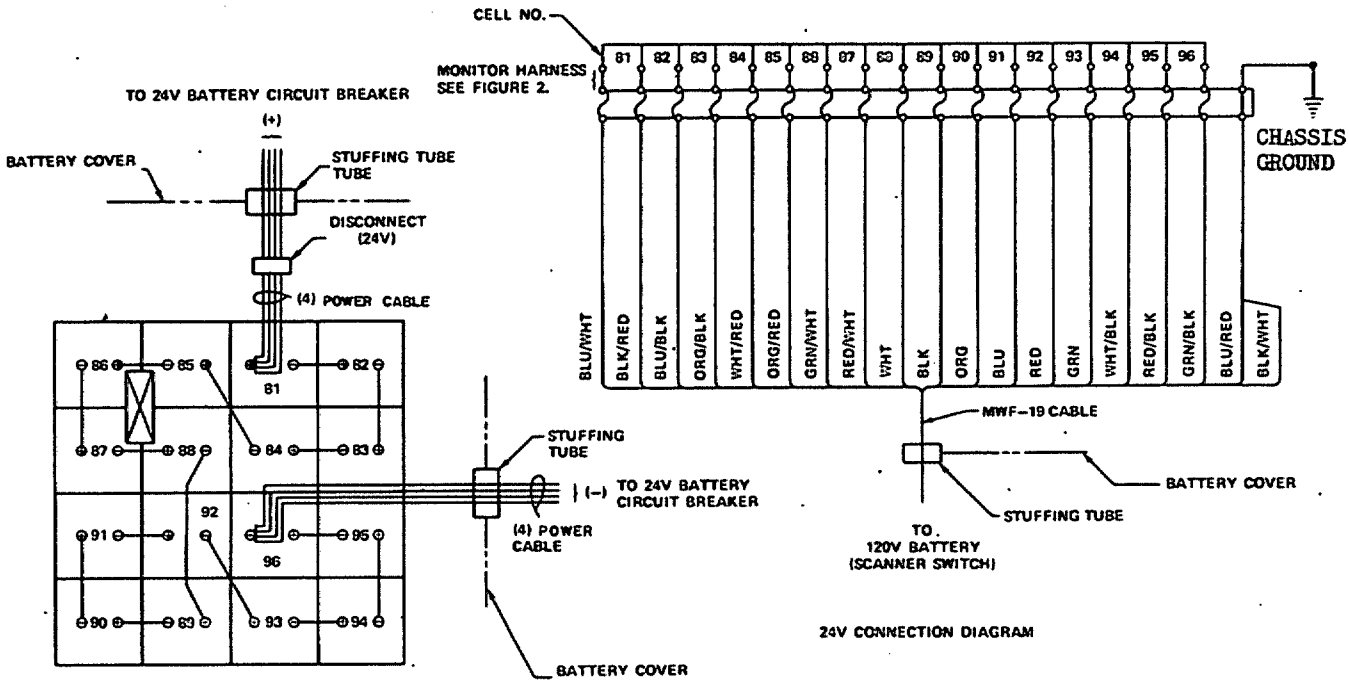
Item no.	Qty reqd	Nomenclature or description
1	16	Cell
2	1	Case, battery (24V)
3	1	Cover, battery (24V)
4	64	Spacer, cell
5	A/R	Wedge, 1.6 mm thk
6	A/R	Wedge, 3.2 mm thk
7	1	Gasket, cover
8	1	Tube, stuffing
9	2	Tube, stuffing
10	2	Cover, manhole
11	1	Cover, manhole
12	2	Gasket, manhole
13	1	Gasket, manhole
14	16	Pad, support
15	54	Spacer, gasket
16	1	Disconnect, bus bar
17	54	Lockwasher - 16 mm (CRES)
18	64	Locknut, 1/2 - 13UNC (NI-CU)
19	10	Nut, hex jam, 1/2 - 13UNC (CRES)
20	2	Plug, drain 3/4 NPT
21	54	Bolt, hex hd 5/8 - 11UNC x 50 mm LG
22	2	Bolt, hex hd (CRES 303) 1/2 - 13UNC x 32 mm LG
23	8	Power cable
24	40	Strap, tie down - self locking
25	A/R	Wire elec -C - 16AWG (blk)
26	1	Cable, MWF-19 or equal
27	A/R	Tape, elec (vinyl) 19 mm wide
28	64	Spacer, gasket
29	2	Lockwasher, 13 mm (CRES)
30	4	Flatwasher, 13 mm (CRES)
31	16	Disc, I.D. (81-96)

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NOTE: DIMENSIONS ARE IN MILLIMETERS. THREADED HARDWARE DESIGNATION IS UNIFIED THREAD STANDARD.

FIGURE 3. Battery assembly (key to figures 1 and 2).

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No. req'd cable or wire	From	To	Length ± 175 mm
(4) Power cable	Stuffing tube	Disconnect	As required to allow cover to lift one meter
(4) Power cable	Disconnect	Cell 81(+)	As required to allow cover to lift one meter
(4) Power cable	Stuffing tube	Cell 96(-)	As required to allow cover to lift one meter
(4) Power cable	Stuffing tube	24 volt breaker (+)	4.3 meters
(4) Power cable	Stuffing tube	24 volt breaker (-)	4.3 meters
(1) MWF-19	Stuffing tube	Cells 81 - 94	As required to allow cover to lift one meter

FIGURE 4. Battery assembly (wiring diagram).

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